



Data-Driven Organizations

AWS Academy Data Engineering

Introduction

Data-Driven Organizations

Module objectives

This module prepares you to do the following:

- Compare and contrast how data analytics and artificial intelligence and machine learning (AI/ML) are applied to data-driven decision-making.
- Name and describe each layer in a data pipeline.
- List ways in which data is acted upon as it passes through a pipeline.
- List responsibilities for data engineer roles and data scientist roles in processing data through a pipeline.
- Define three modern data strategies that influence how you build your data infrastructure.

Module overview

Presentation sections

- Data-driven decisions
- The data pipeline – infrastructure for data-driven decisions
- The role of the data engineer in data-driven organizations
- Modern data strategies

Knowledge checks

- Online knowledge check
- Sample exam question

Data-driven decisions

Data-Driven Organizations



"Through 2026, organizations will accelerate their investments in data and analytics services by 45 percent to become more data driven and digital."

Gartner

Forecast Analysis: Data and Analytics Services, Worldwide,
Twiggy Lo, March 2022

How do you decide...

- Which restaurant near me serves the best Thai food?
- Where can I find a used child's bicycle?
- Which starters should I pick for my fantasy football team?
- Which type of job should I pursue?
- What should I pay for this home?

How do organizations decide...

- Which of these customer transactions should be flagged as fraud?
- Which webpage design leads to the most completed sales?
- Which patients are most likely to have a relapse?
- Which type of online activity represents a security issue?
- When is the optimum time to harvest this year's crop?

Fueling decisions with data science

Data analytics

- Is the systematic analysis of large datasets (big data) to find patterns and trends to produce actionable insights
- Uses programming logic to answer questions from data
- Is good for structured data with a limited number of variables

AI/ML

- Is a set of mathematical models that are used to make predictions from data at a scale that is difficult or impossible for humans
- Uses examples from large amounts of data to learn about the data and answer questions
- Is good for unstructured data and where the variables are complex

Example: Identify pictures of dogs

Data analytics approach

Based on a set of defined features, which of these pictures are dogs?



AI/ML approach

Which of these images have the same features as the pictures that I have seen that were labeled as dogs?



Business example: Customer relationship management

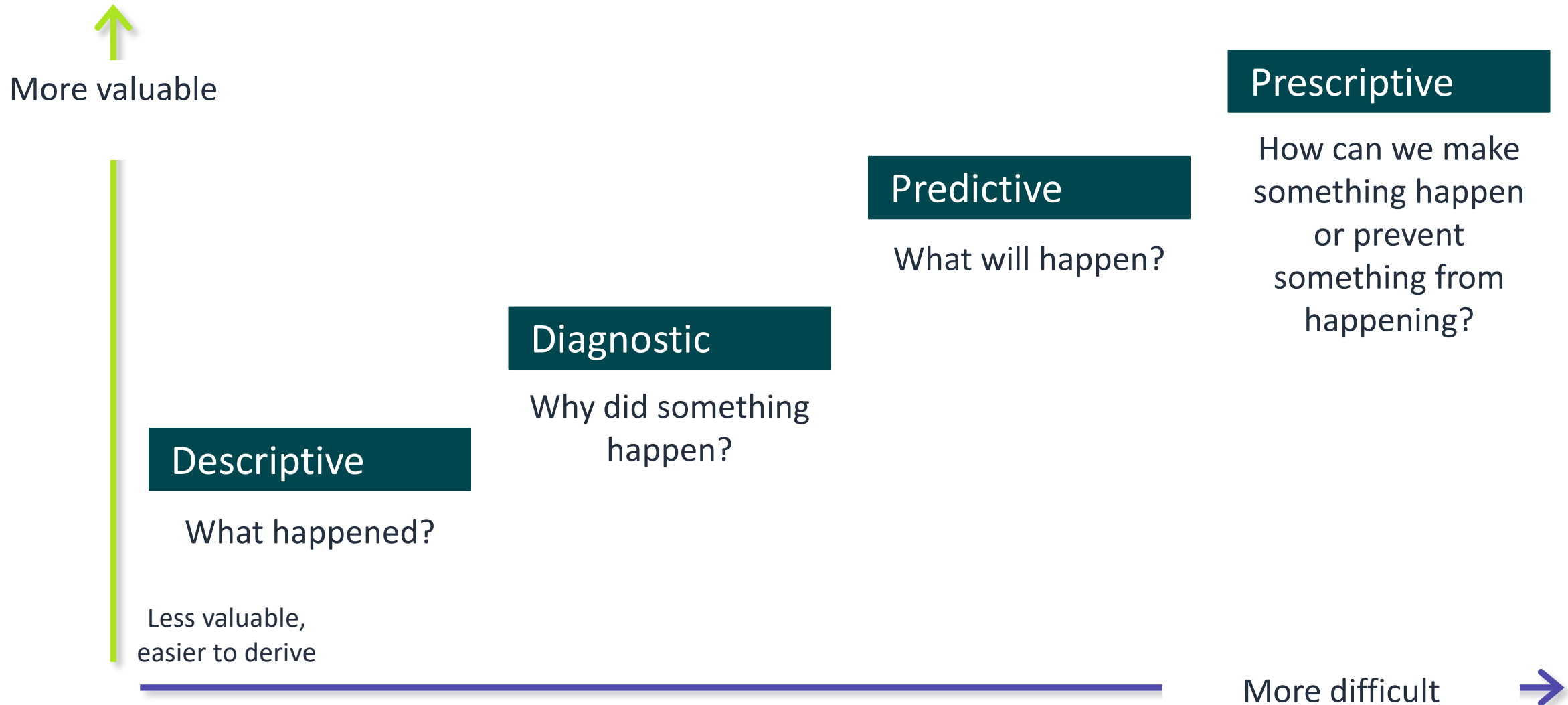
Data analytics approach

- A retail business analyzes total revenue per customer and segments customers into categories based on spending.
- The segmentation might be used to give a higher level of customer service to customers who spend more.

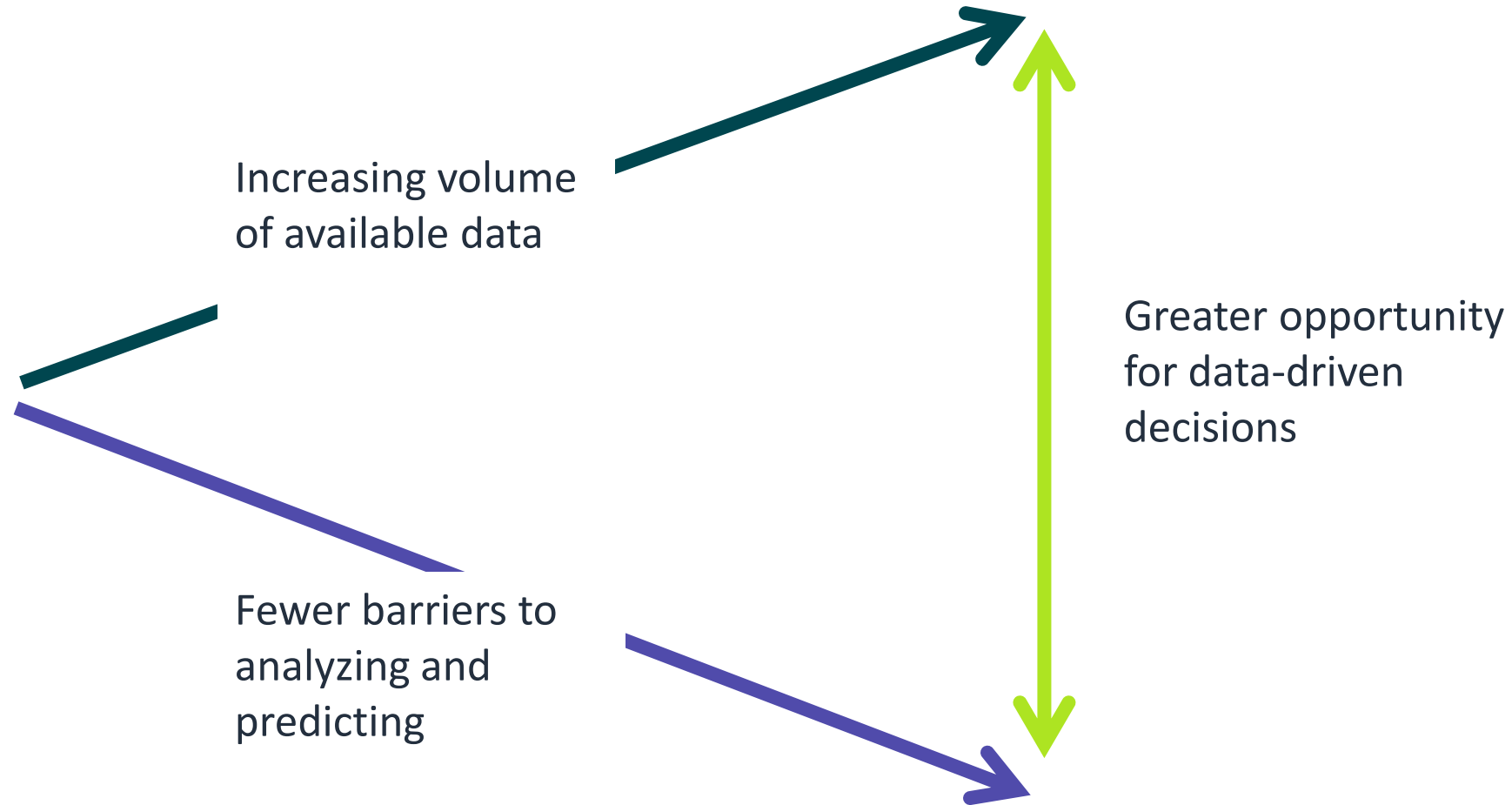
AI/ML approach

- A retail business uses AI/ML to analyze customer churn (why and how often customers come and go).
- AI/ML might uncover factors that influence churn so that the business can make changes to better retain customers.

More valuable insights are more difficult to derive



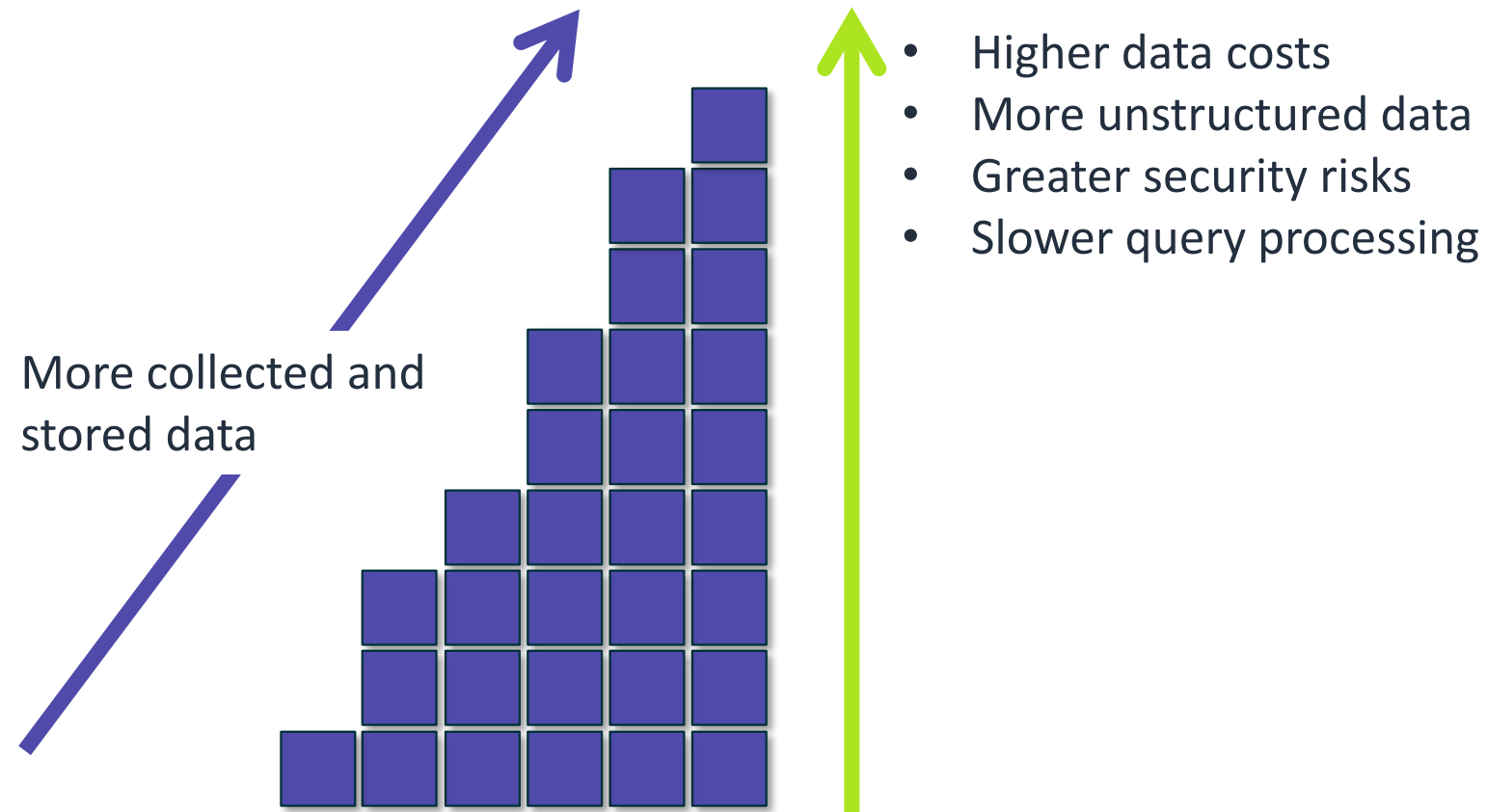
More data + fewer barriers = more data-driven decisions



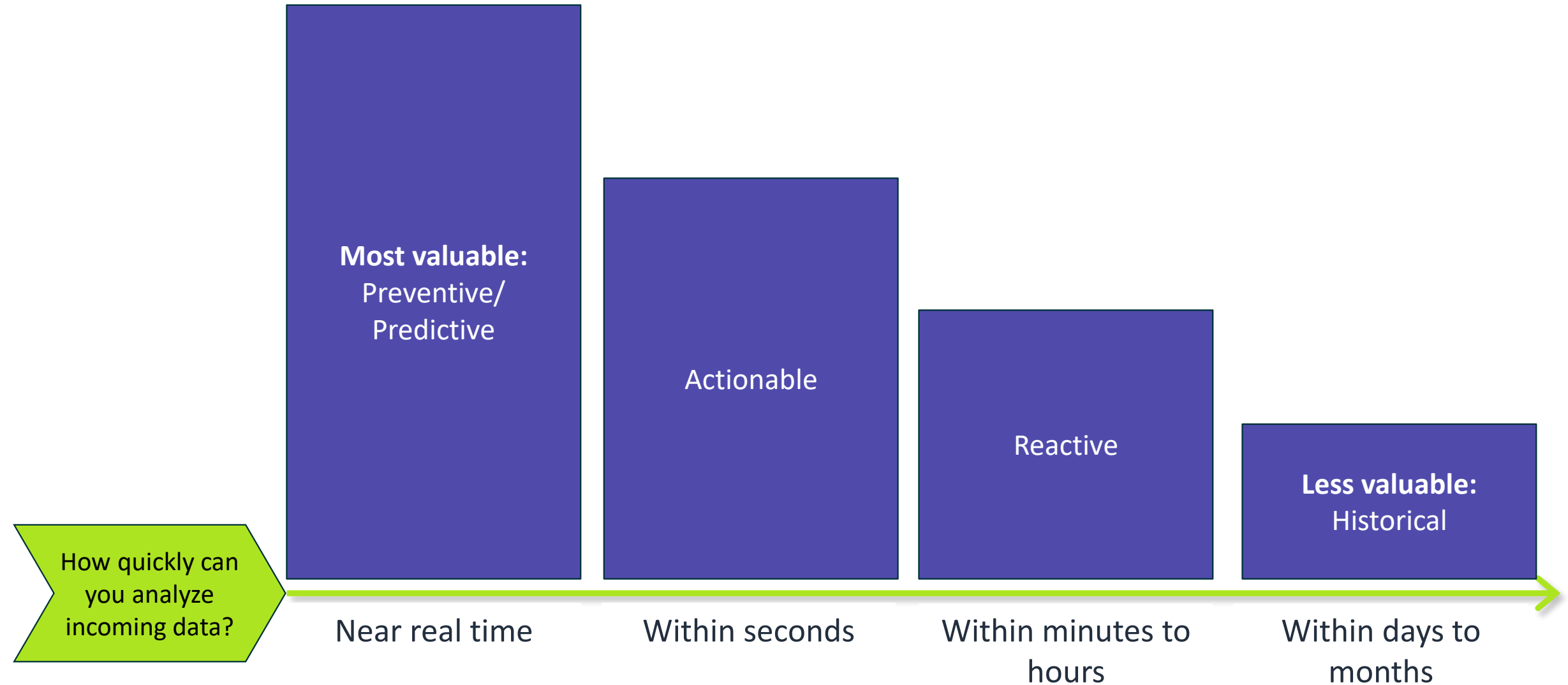
More data science in daily life

- You are shopping for shoes online and start to see a lot of shoe-related advertisements.
- You watch a movie or listen to a song on a streaming platform and begin to get recommendations for movies or music you might also like.
- You order pizza online and are kept up to date about each step in the preparation and delivery process.
- You use your credit card outside of your usual geographic area and get additional fraud alerts from your bank.
- Your navigation app on your phone alerts you to a traffic jam.

More data doesn't necessarily equal more value



Data becomes less valuable for decision-making over time



The trade-offs of data-driven decisions



Cost

- How much should you invest to go faster or predict more accurately?
- How much incremental improvement justifies additional cost?



Speed

- How quickly do you need an answer?
- Can you sacrifice accuracy for speed?



Accuracy

- How accurate does the prediction need to be?
- Does waiting for a better answer outweigh answering more quickly?

Key takeaways: Data-driven decisions



- Data-driven organizations use data science to make informed decisions.
- Data analytics relies on programming logic and tools to arrive at predictions from data. This approach is good for structured data with a limited number of variables.
- AI/ML can learn to make predictions from examples in data. This approach is good for unstructured data and where the variables are complex.
- A large increase of available data, coupled with a decrease in the cost of relevant technology, has increased the opportunity for data-driven decisions.

The data pipeline – infrastructure for data-driven decisions

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A data pipeline in its simplest terms



Collect data



Store and process
data



Build something
useful with data

Work backwards to design your infrastructure



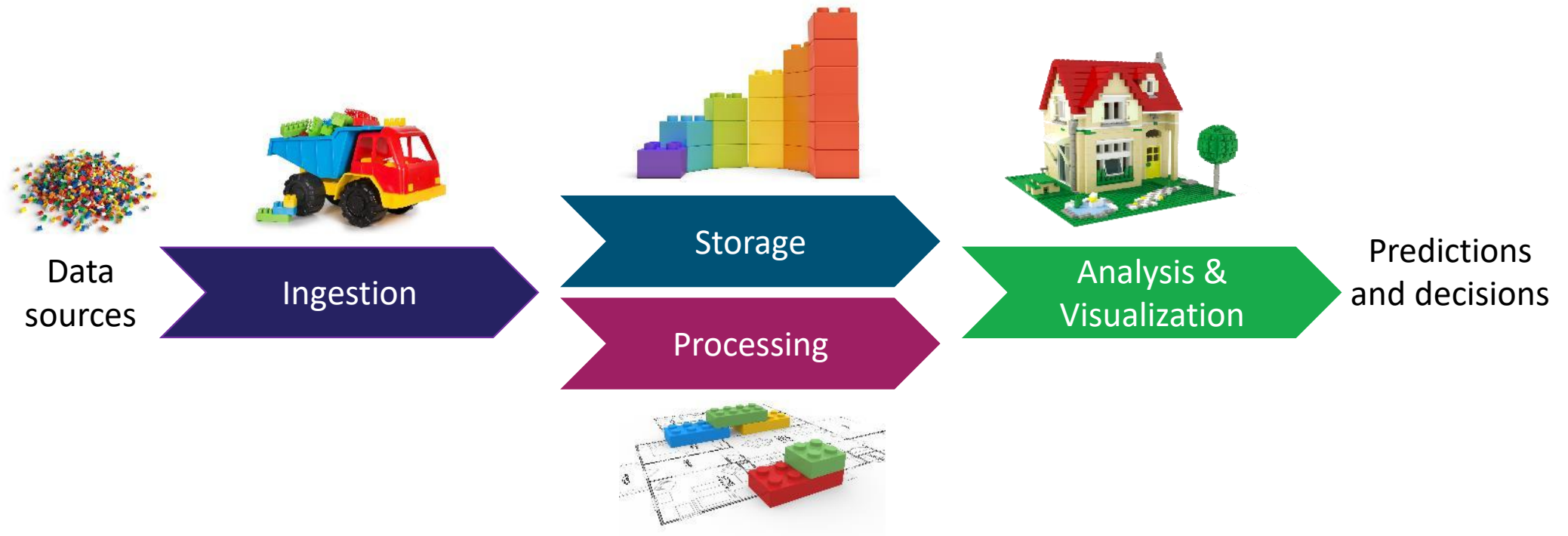
2. What data do you need to support this?

1. What decision are you trying to make?

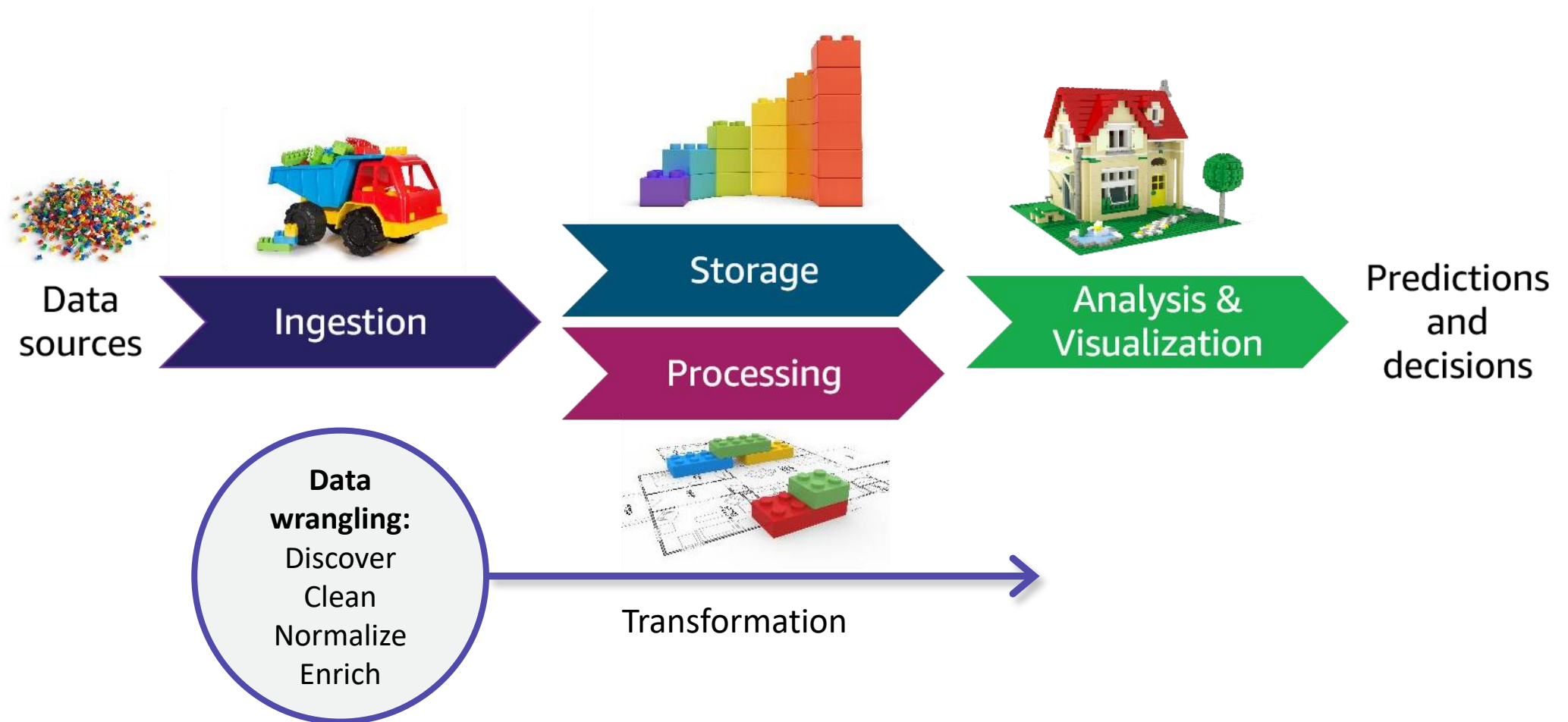


Weigh the trade-offs of cost, speed, and accuracy.

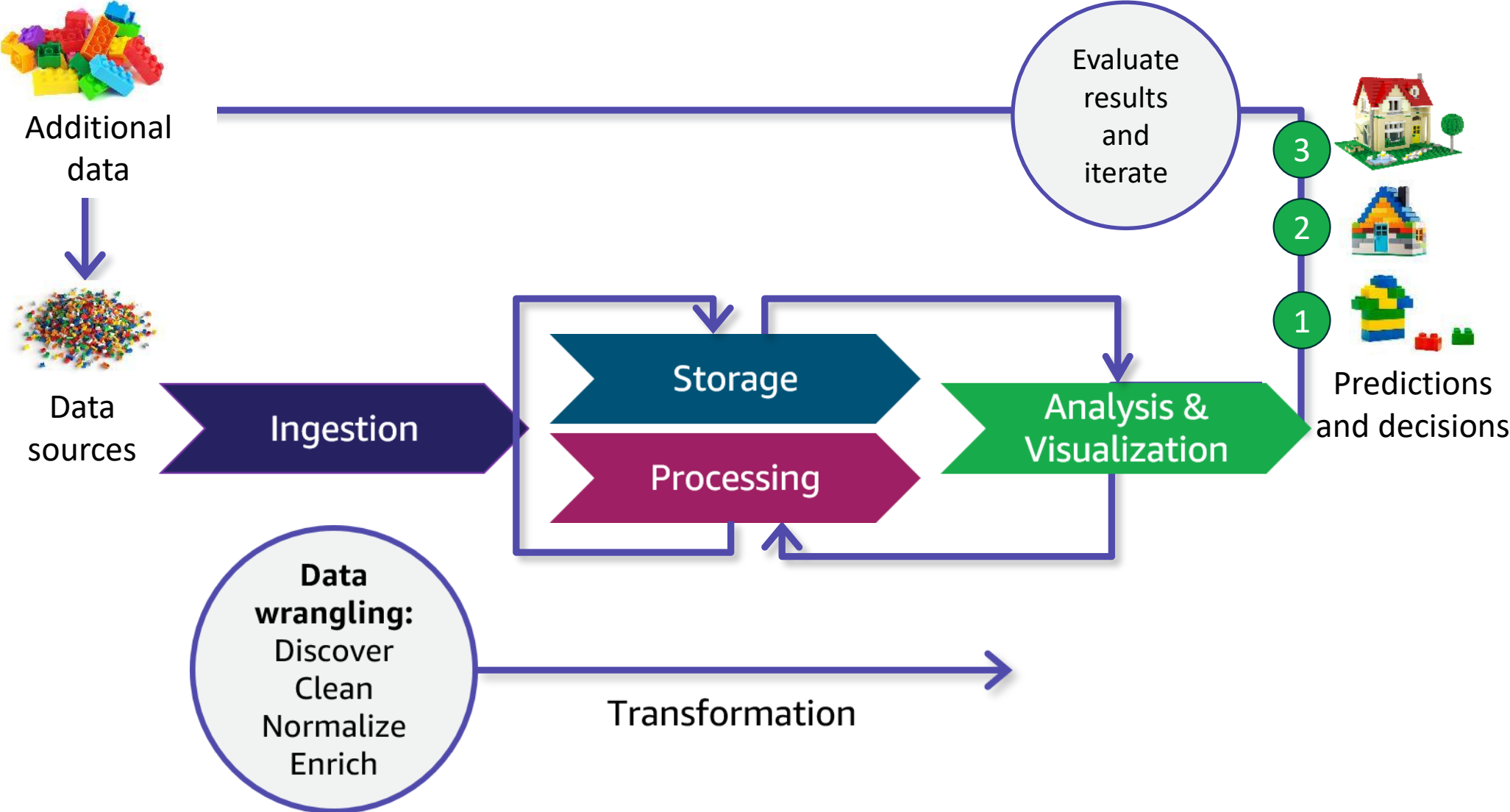
Layers of the pipeline infrastructure



Actions taken with data in the pipeline



Iterative processing through the pipeline



Key takeaways: The data pipeline – infrastructure for data-driven decisions



- A data pipeline provides the infrastructure for data-driven decision-making.
- When designing a pipeline, start with the business problem to be solved and work backwards to the data.
- The pipeline includes layers to ingest, store, process, and analyze and visualize data passing through it.
- Data wrangling refers to how data is acted upon as it passes through the pipeline. Tasks include discovery, cleaning, normalization, transformation, and augmentation.
- Data is processed iteratively to evaluate and improve upon results.

The role of the data engineer in data-driven organizations

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Common data pipeline questions



Data
engineer

- Does the organization have the data that addresses the need? Where is the data stored and in what format?
- Will I need to combine data from multiple sources?
- What is the type and quality of data? What will be the source of truth?
- What are the security requirements for this data? Who needs access to it and in what state?
- What types of mechanisms are needed to transfer the data from its locations into the pipeline?
- How much data is there, and how frequently is it updated or processed?
- How important is speed when data is requested?



- What can the data tell me?
- How will I evaluate the results?
- What kind of visualization do I need?
- Which formats and tools are the analysts familiar with?
- Do I need a big data framework?
- Which type of AI/ML models fit?
- What is the simplest way to implement AI/ML?



Data
scientist

Key takeaways: The role of the data engineer in data-driven organizations

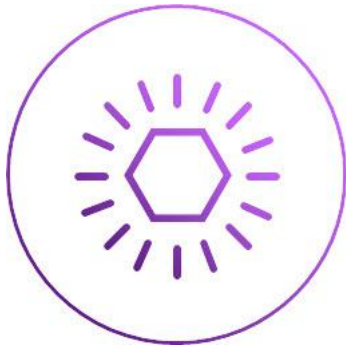


- Both data scientist roles and data engineering roles work with the data pipeline, and tasks might be performed by someone in either role.
- Data engineering is primarily focused on the infrastructure that the data passes through while the data scientist works with the data in the pipeline.
- To build the right pipeline, you need to ask questions about the desired outcomes and the data and then iterate on your answers as you learn more.

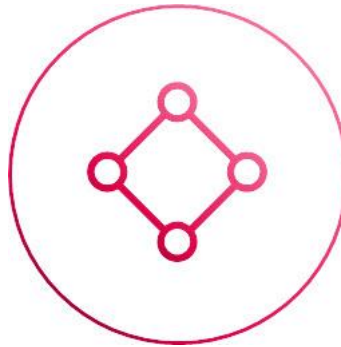
Modern data strategies

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Three-pronged strategy to build data infrastructure



Modernize



Unify

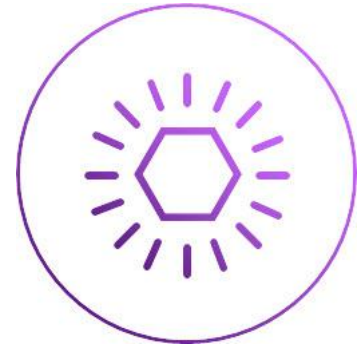


Innovate

Modernize

Increase agility and reduce undifferentiated lifting

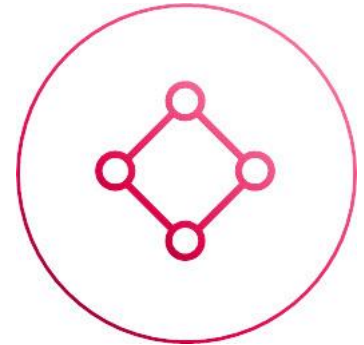
- Move from on premises to cloud-based services.
- Migrate to purpose-built tools and data stores.
- Build loosely coupled pipelines.



Unify

Create a single source of truth

- Break down data silos.
- Democratize access.
- Equip users with tools to visualize their own insights.
- Use a data lake, and run queries directly on data.
- Support simplified governance and movement between the data lake and purpose-built stores.



Innovate

Use AI and ML to discover new insights faster

- Move from reactive to proactive decision-making.
- Incorporate AI/ML into decision-making, and tap into new insights in vast amounts of unstructured data.
- Take advantage of cloud services with AI/ML features that democratize who can use ML.





"Being a data-driven organization means culturally treating data as a strategic asset and then building capabilities to put that asset to use – not just for big decisions but also for everyday action on the front line."

Ishit Vachhrajani, AWS Enterprise Strategist

Key takeaways: Modern data strategies



- Organizations that want to become data driven should modernize, unify, and innovate with their data infrastructures.
- Modernizing is about moving to cloud-based infrastructures and purpose-built services to reduce administrative and operational effort.
- Unifying is about creating a single source of truth for data and making the data available across the organization.
- Innovating is about looking for new ways to find value in the data—specifically, applying AI and ML to find new insights.

Module wrap-up

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Module summary

This module prepared you to do the following:

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Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions that are based on material from the slides and slide notes.
- You can retake the knowledge check as many times as you would like.

Sample exam question

A traditional retailer with a relatively small online presence recently decided that they need to expand their online presence to increase sales. After a major marketing push and investing in IT infrastructure to support increased online transactions, quarterly results show that online sales results are far below expectations. The business wants to find ways to improve the poor online sales performance.

Which action represents an appropriate first step to address the request in a data-driven way?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- **Improve online sales**
- Appropriate **first step**
- **Data-driven** way

Sample exam question: Response choices

A traditional retailer with a relatively small online presence recently decided that they need to expand their online presence to increase sales. After a major marketing push and investing in IT infrastructure to support increased online transactions, quarterly results show that online sales results are far below expectations. The business wants to find ways to **improve the poor online sales** performance.

Which action represents an appropriate **first step** to address the request in a **data-driven way**?

Choice	Response
A	Develop an ML model that analyzes customer comments to determine where customers are unsatisfied.
B	Use A/B testing to provide some customers with a new UI that is designed to simplify the purchasing process. Then, compare sales results from the old and new UIs.
C	Use a business intelligence (BI) tool to analyze sales data for the previous two quarters. Then, generate a hypothesis as to why the online sales are not in line with expectations.
D	Analyze the total revenue per customer, and create customer segments that receive different types of offers based on their segment.

Sample exam question: Answer

The correct answer is C.

Choice	Response
C	Use a business intelligence (BI) tool to analyze sales data for the previous two quarters. Then, generate a hypothesis as to why the online sales are not in line with expectations.

Thank you



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